Reg. No. | | | | | | | | | | |

[14ME652]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE–2019

QUALITY ASSURANCE AND RELIABILITY ENGINEERING

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions, choosing at least TWO questions from each part.
2. Missing data, if any, may be suitably assumed.
3. Use of statistical tables permitted.

PART A

1. a. Define the term quality, control charts and process capability with suitable examples. **06**

   b. Briefly explain 'variation in process' and list the various control charts available for the variables with control limits. **08**

2. a. Distinguish between c-Chart and u-Chart and list various Control Charts available for attributes with control limits **06**

   b. In a production unit 15 lots of 400 items each are subjected to 100% inspection and the number of defectives found in each sub group is as shown in the table below: **08**

Lot Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
No. of Defectives	2	5	0	14	3	0	1	0	18	6	6	0	3	0	6

   i. Compute process average for np chart

   ii. Compute trial control limits for np chart

   iii. Draw np chart

   Draw conclusions

3. a. Explain briefly the characteristics of OC curve and list the various types of sampling plans. **06**

   b. The Manufacturing organization is using a double sampling plan with N = 4,000, n1 = 150, c1 = 4, n2 = 200, c2 = 7 and p' = 0.015. Calculate ATI, AOQ of the above plan. **08**

4. a. Define and explain with suitable examples producer's risk and consumer's risk **08**

   b. Write a short notes on ; **06**

   i. AQL

   ii. AOQL and

   iii. LTPD

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PART – B

5. a. Define reliability. 02

b. Write short notes on failure rate curve. 04

c. A system consists of five components connected as shown in figure 5.c with given values of reliabilities. Determine the System reliability? 08

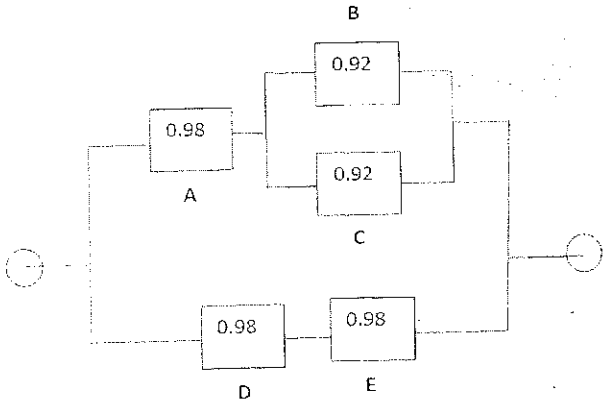

Fig. 5 c

6. a. A mechanical equipment is subjected to a maintenance time of 45 minutes. What is the probability that it will meet the specification if MTTR is 0.262 hours? 06

b. What are the steps involved in designing for higher Reliability. 08

7. a. Explain briefly; 08

i. Constant hazard model and

ii. Linear hazard model

b. Explain fault tree technique with respect to reliability along with suitable example. 06

8. a. Explain component redundancy versus unit redundancy. 06

b. Write a short note on; 08

i. Mixed redundancy

ii. Redundancy optimization